YUZHUO XIAO

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EDUCATION

GuiZhou University (GZU) *June 2026 (expected)*
B.E. in Computer Science (Advanced Class)

RESEARCH INTERESTS

My research interests focus on video generation. I am also interested in Multimodal evaluation and Natural Language Processing.

PUBLICATIONS

[1] Y. Xiao, Z. Han, Y. Wang, and H. Jiang, *Xfacta: Contemporary, real-world dataset and evaluation for multimodal misinformation detection with multimodal llms*, 2025. arXiv: 2508.09999 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2508.09999.

RESEARCH EXPERIENCE

XFacta: Contemporary, Real-World Dataset and Evaluation for Multimodal Misinformation Detection with Multimodal LLMs [1] under review
Yuzhuo Xiao*, *Zeyu Han*, Yuhan Wang, Huaizu Jiang* NEU (remote)

- Project leader
- Project: XFACTA, a contemporary real-world multimodal misinformation detection project, focusing on dataset construction and systematic analysis of evidence retrieval and reasoning in MLLMs.
- Contributions: Built a large-scale contemporary real-world multimodal misinformation dataset and performed systematic analysis of evidence retrieval and reasoning strategies for MLLM-based detection.
- Outcomes: Achieved state-of-the-art performance on the benchmark and enabled continuous dataset growth via a validated detector-in-the-loop pipeline.

Human-Centric Video Generation Research ongoing
Supervisors: Huaizu Jiang NEU (remote)

- Project leader
- Project: Studying how fast and large-magnitude human motion affects spatiotemporal consistency in human-centric video generation models.
- Contributions: Conducted comparative studies on representative human-centric video generation models; designed controlled experiments with fixed prompts and varying motion descriptors; analyzed model behavior under complex motion patterns.
- Status: Ongoing; problem formulation and experimental design completed.

TECHNICAL EXPERIENCE

RecSys Challenge 2025: Universal User Behavior Representation Competition Project

- Project: Universal behavioral profile learning for multi-task recommendation and user modeling.
- Contributions: Manually designed and implemented behavior-based statistical features from large-scale user interaction logs; explored feature construction strategies focusing on behavior frequency, temporal dynamics, and cross-action relationships; conducted systematic ablation studies to analyze the impact of feature quality versus model complexity.

- Outcomes: Built a robust and reproducible feature-driven user representation pipeline that consistently improved performance across multiple public and hidden downstream tasks under changing data distributions.

Parameter-efficient Fine-tuning for Machine Translation Course Project

- Project: An end-to-end machine translation project for low-resource Bengali–English translation using parameter-efficient fine-tuning.
- Contributions: Designed and implemented a LoRA-based fine-tuning pipeline on mBART-large; performed data cleaning, filtering, and sampling on large-scale parallel corpora; trained models on 300K sentence pairs.
- Outcomes: Achieved over 200% relative BLEU improvement over the pretrained mBART baseline on held-out news test sets, with consistent gains in ChrF.

Digital Image Processing: Classical Methods and Deep Learning Course Project

- Project:Classical and Deep Learning–Based Image Processing System
- Contributions: Implemented edge detection, frequency-domain processing, color space conversion, image segmentation, and object detection algorithms. Applied U-Net for pixel-level defect segmentation and compared with traditional methods.
- Outcomes:Built end-to-end image processing pipelines using OpenCV and PyTorch.

Parkinson's Disease Classification from Speech Data Course Project

- Project: Machine Learning–Based Automatic Diagnosis of Parkinson's Disease from Speech
- Contributions: Focused on speech feature modeling, manually implemented Logistic Regression, Random Forest, and SVM algorithms and applied them to automatic Parkinson's disease diagnosis.
- Outcomes: Established a reproducible experimental framework.

ACADEMIC TRAINING AND SUMMER PROGRAMS

AI & Robotics Training Program - SenseTime Summer 2023

- Completed a training program on deep learning and computer vision, with hands-on experience in dataset curation, model training, and experimental evaluation. Built a custom visual dataset and trained classification models using standard deep learning pipelines. Gained practical experience in perception systems through integrating vision sensors for embodied tasks.

Multimodal AI Summer Course (Guest Lectures by USTB) Summer 2024
Instructors: Dr. Dandan Song (USTB) and Prof. Ruizhang Huang

- Participated in a summer course on multimodal learning, covering vision-language model architectures, multimodal alignment, and representative multimodal frameworks. Completed a final project involving the deployment and adaptation of a Multimodal LLM (MiniCPM) for downstream tasks, strengthening practical understanding of multimodal systems.

AWARDS AND HONORS

DaBeiNong First-Class Scholarship	*Summer 2025*
Guizhou University First-Class Scholarship	*Summer 2023*
Guizhou University "Three Good Student" Award	*Summer 2023*

SKILLS/HOBBIES

Programming Languages	Python, C/C++, Bash
Machine Learning Tools	Pytorch, Tensorflow, Sklearn, Pandas, Numpy, OpenCV
Hobbies	birding and dancing